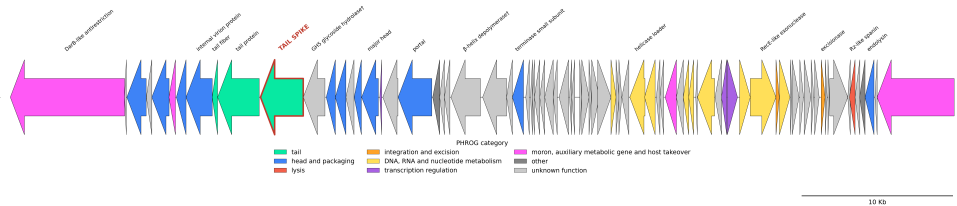
Engineering the part that picks the host

Which gene, what it binds, where evolution comes in



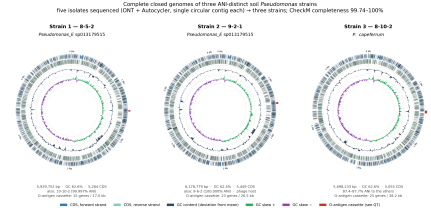
Phage 9-2-1 Φ8 — 62 kb, 80 genes — and three self-isolated *Pseudomonas* strains

Chun-Yu (Alex) Chen | Phagenomics, Claremont CA | PI: J. Cesar Ignacio-Espinoza | 2026-08-21

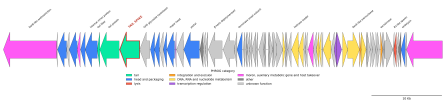
We want to design the part, not just describe the phage

- Phages hunt bacteria, and one phage usually infects **one strain** and ignores the next one over. Matching is done by **trial and error** today.
- Why now: **[AMR figure + source — Alex]**. A phage is only useful if it hits the strain that is actually making someone sick.
- Our goal is a **part you can re-aim**: rebuild the receptor-binding protein so the same phage attacks a different strain.
- **We work on our own material** — local soil and water → *Pseudomonas* hosts and their phages, sequenced in house.

Tonight: 5 isolates → **3 strains**, and one complete phage genome



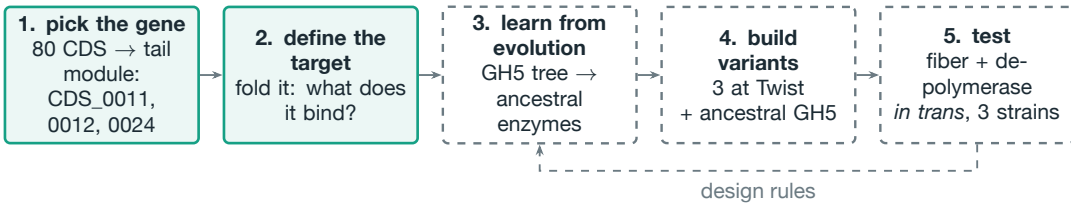
three closed host genomes, 5.7–6.2 Mb



the phage: 62 kb, 80 CDS, linear + 77 bp DTR

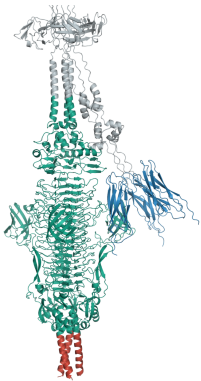
The design loop — and where we actually are

Before you can redesign the targeting part: which gene is it, and what must it recognise?



solid = has data dashed = in progress / proposed | the next two slides are steps 1 and 2

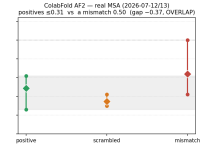
Steps 1–2: a β-helix depolymerase — the target is a sugar



CDS_0011 trimer, 941 aa × 3
grey shaft • blue anchor

teal body • red foldon

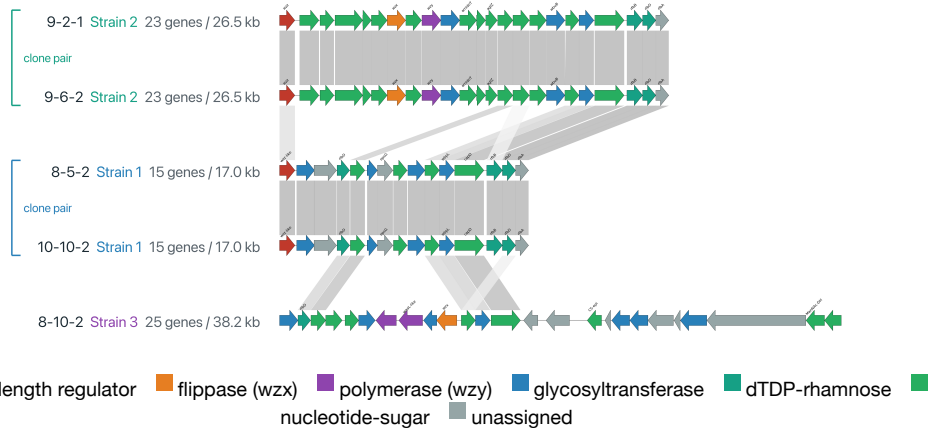
- Right-handed β**-helix**, one rung every **15–18 residues** — the signature of a polysaccharide depolymerase. **This part cuts sugar**, so its target is the surface sugar, not a protein.
- The domain map is what makes it **engineerable**: pLDDT 88–94, and the body (res 301–941) is rigid to **0.44–0.99 Å** across five models — that boundary is exactly where our construct starts (res 305).
- The protein–protein hypothesis was tested first **and failed**:



positives ipTM $\leq$ 0.31 vs a mismatched pair 0.50 — no signal

The target is different in every strain — that is the handle

Same `ihfB`-flanked slot, different cargo (grey ribbons = homology)



4 of 45 clusters shared by all three — generic sugar chemistry only, no shared assembly machinery. Control: clone pairs 100.0% identical.

Where the loop stands

Conclusion

We know **which gene** to engineer and **what it must recognise**: the tail-spike depolymerase, against a strain-specific O-antigen — and our three strains give three different targets.

Limitation

Steps 3–5 are **not results yet**: the GH5 ancestral reconstruction is still being built, the constructs are not expressed, and the cross-strain plaque assay was run July 4 but the data is not in our hands. It also assumes affinity tracks infection. No serotyping tool exists for this taxon, so we cannot name an O-type.

Next three months

Recover that readout; express the three constructs; reconstruct ancestral GH5 enzymes and test fiber + depolymerase *in trans* on the three strains. The tail spike also has a 15,775-sequence alignment — enough for a conservation profile on the structure.



only Strain 2 has actually been challenged; the grey cells are predictions, and the July 4 assay reads them out

1. **pick the gene** — done
2. **define the target** — done
3. **ancestral GH5 enzymes** — PI building the tree
4. **build** — 3 constructs at Twist
5. **test** — *in trans*, waiting on the July 4 readout

We now know which part to rebuild and what it has to grip — the ancestral variants are the next lever.

Backup

not shown unless asked

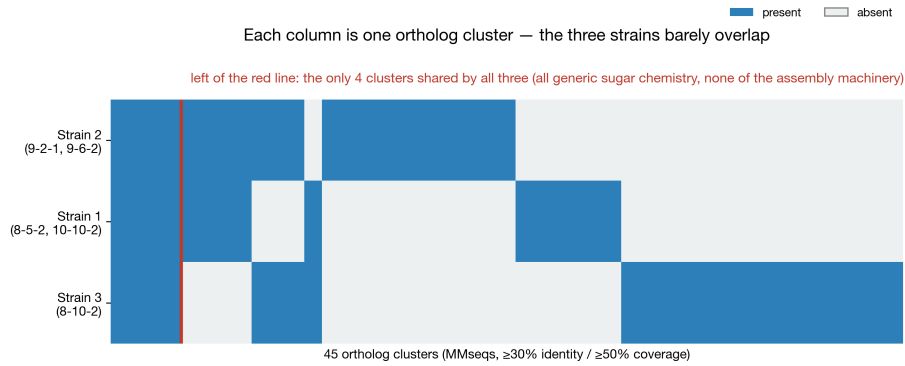
Backup 1 — gene selection and the constructs

- Structural module, in canonical podovirus order: portal (CDS_0020) → major capsid (0017) → adaptors → **tail spike (0011)** → tail (0010) → fiber (0009).
- **CDS_0024 is a second β-helix depolymerase** (rung 22–24) that Pharokka called “hypothetical”; the de-novo RBP classifier found it at score 0.998.
- **CDS_0012 is a GH5 glycoside hydrolase** — a separate protein, not a domain of the tail spike; all four AF3 templates are GH5 endoglucanases.
- Ordered from Twist 2026-07-27, all in pET-28a(+) with N-terminal His6: CDS_0011 res **305–941** (the rigid body), CDS_0010 res 105–917, CDS_0012 res 117–477.
- CDS_0010 is an **unknown protein** — the glycosidase call was withdrawn 2026-07-29; it is being expressed for functional characterisation.

Backup 2 — how the locus was pulled, how orthology was called

- Anchor, identical in all five genomes: 30S ribosomal protein S1 $\rightarrow$ `ihfB` $\rightarrow$ LPS assembly protein A $\rightarrow$ MBL-fold hydrolase $\rightarrow$ **cassette starts**. `ihfB` is the canonical upstream flank of the *Pseudomonas* OSA cluster in the literature.
- Orthology = MMseqs2 clustering, $\geq$30% identity / $\geq$50% coverage. **Product names cannot be used**: the chain-length regulators of Strain 1 and Strain 2 carry different Bakta labels but are homologous by sequence.
- Locus sizes: Strain 2 = 23 genes / 26.5 kb • Strain 1 = 15 / 17.0 kb • Strain 3 = 25 / 38.2 kb.
- The cross-strain best-hit numbers (9/23, 10/23, 6/15) are one-directional best hits at a looser cutoff and are not meant to reconcile with the 45/4 cluster counts.

Backup 3 — cluster matrix, 45 clusters × 3 strains



Backup 4 — limitations in full

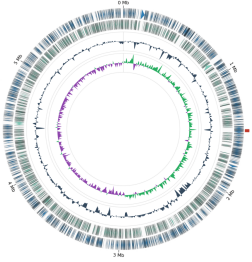
- **Variable $\neq$ specificity-determining.** An earlier plan to read specificity off metagenomic variable regions was rejected in our own review: metagenome sequences carry no host labels, so specificity-determining positions cannot be identified that way. The 15,775-sequence alignment (tail spike, CDS_0011) supports a **conservation profile** — not "these residues set the host range". The ancestral reconstruction the PI is building is a separate thing: a tree over **all GH5 sequences** (CDS_0012), whose ancestral enzymes get expressed and challenged *in trans*, under the stated caveat that affinity $\sim$ binding $\sim$ infection.
- **Do not say Strain 1 lacks a flippase or polymerase** — Bakta assigns no Pfam/InterPro to those two genes; confirming them needs InterProScan.
- **The sugar structure has not been measured** — gene content is not structure; that needs LPS extraction and mass spectrometry.
- **The β-helix has no named structural homolog yet** — no foldseek or DALI run.

Backup 5 — the three host genomes

Complete closed genomes of three ANI-distinct soil *Pseudomonas* strains
five isolates sequenced (ONT + Autocycler, single circular contig each) → three strains; CheckM completeness 99.74–100%

Strain 1 — 8-5-2

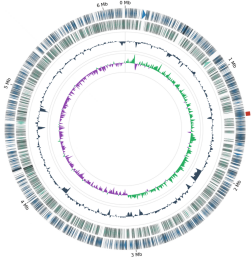
Pseudomonas_E sp013179515



5,929,752 bp · GC 62.6% · 5,264 CDS
also: 10-10-2 (99.997% ANI)
O-antigen cassette: 15 genes / 17.0 kb

Strain 2 — 9-2-1

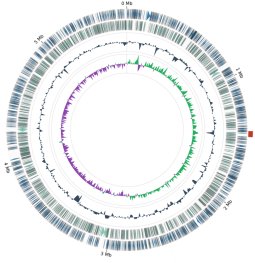
Pseudomonas_E sp013179515



6,178,779 bp · GC 62.4% · 5,449 CDS
also: 9-6-2 (100.000% ANI) · phage host
O-antigen cassette: 23 genes / 26.5 kb

Strain 3 — 8-10-2

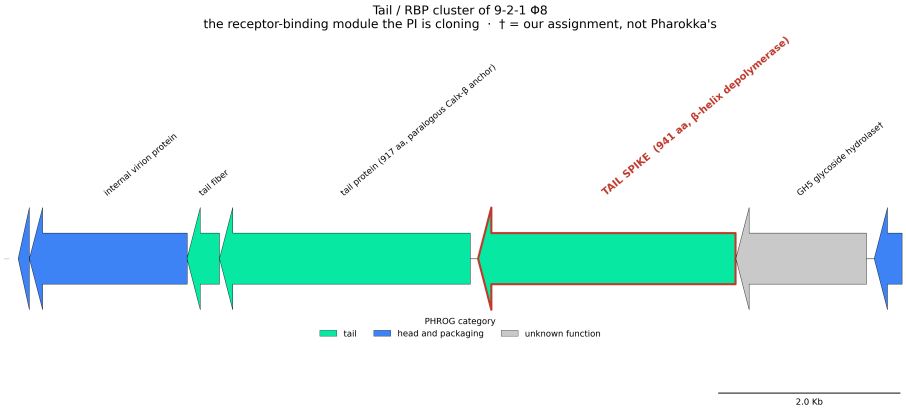
P. capeferrum



5,698,233 bp · GC 62.6% · 5,055 CDS
87.4–87.7% ANI to the others
O-antigen cassette: 25 genes / 38.2 kb

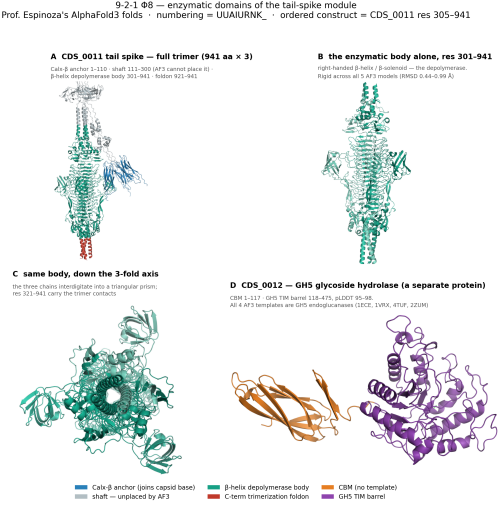
CDS, forward strand · CDS, reverse strand · GC content (deviation from mean) · GC skew + · GC skew − · O-antigen cassette (see Q7)

Backup 6 — phage genome, tail / RBP module



Numbering is the PI's `UUAIURNK_`. The old 0708 genome map is mirrored — do not reuse it.

Backup 7 — enzymatic domains in full (incl. CDS_0012 GH5)



Panel D is CDS_0012 — a **separate protein**, not a domain of the tail spike.